



Faculty Of Engineering
Biomedical Engineering Department

Course Title
Biomedical Electronics

Report title
EEG Project

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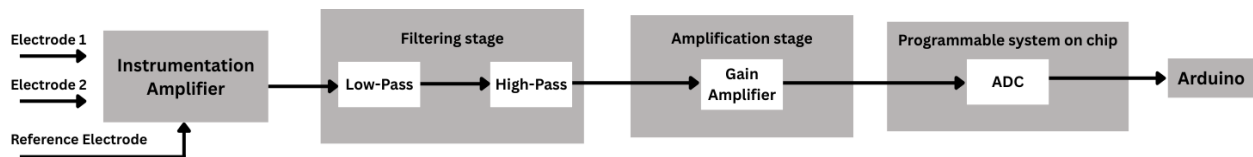
Circuit design

Overview of the Circuit:

This section describes the design and implementation of a single-channel EEG signal acquisition circuit developed as part of an academic course project. The circuit is designed to acquire low-amplitude EEG signals from surface electrodes and condition them for reliable digital conversion and visualization.

The acquisition channel includes an instrumentation amplifier followed by analog filtering stages, including high-pass, low-pass, and notch filters, in order to remove baseline drift, high-frequency noise, and power-line interference. After signal conditioning and amplification, the output is connected to an ADS1115 analog-to-digital converter.

The ADS1115 digitizes the EEG signal and communicates with an Arduino microcontroller via the I²C interface. The digitized data is then transmitted to a MATLAB environment, where the signal is processed and visualized in real time. The system is powered using isolated supply rails to reduce noise and ensure safe operation.



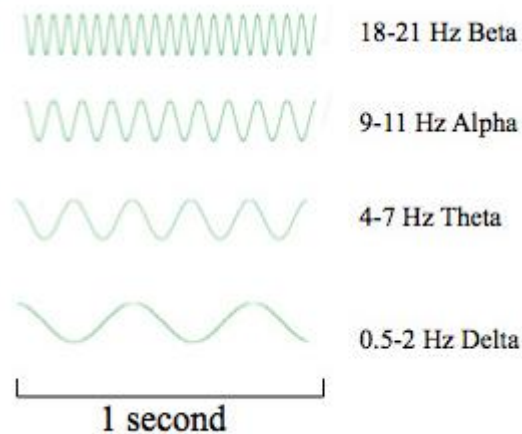
Block Diagram of the Circuit

Filtering

Due to their weak amplitudes, EEG signals are very susceptible to electromagnetic and common mode contamination. Additionally, EEG signals collected with electrodes may contain EMG information from the subject's muscular activity and ECG signals from the polarizing cycles of heart cells. Is it thus important to filter the signal in order to isolate the frequency bands of interests for EEG analysis. The ElectroEncephaloGram is composed of 4 distinct waves ranging between 0.2 and 25 Hz;

1. The beta waves (18-21Hz)
2. The alpha waves (9-11Hz)
3. The theta waves (4-7Hz)
4. The delta waves (0.5-2Hz).

Therefore, the bandwidth filter was designed to range from 0.3 to 35Hz to preserve all relevant EEG information.



Frequencies of EEG waves

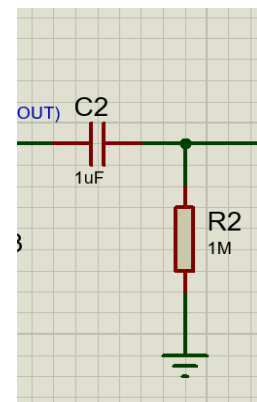
High pass

A first-order high-pass filter is implemented before the 1st amplification stage. This filter is used to remove the DC offset and low-frequency baseline drift present in the acquired EEG signal. It is realized using a passive RC network and ensures that only the relevant AC components of the EEG signal are passed to the subsequent amplification stage.

The cutoff frequency is calculated by this equation

$$F = \frac{1}{2\pi RC}$$

The cutoff Frequency is then calculated to be = 0.16 Hz



1st Order High-Pass

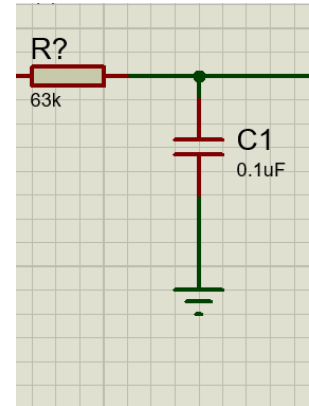
Low pass

A first-order low-pass filter is implemented before the 2nd amplification stage. This filter limits high-frequency noise and defines the upper bandwidth of the EEG signal before further amplification. The filter is implemented using a passive RC network and helps improve the signal quality by attenuating unwanted high-frequency components.

The cutoff frequency is calculated by this equation

$$F = \frac{1}{2\pi RC}$$

The cutoff Frequency is then calculated to be = 25 Hz



1st Order Low-Pass Filter

Amplification

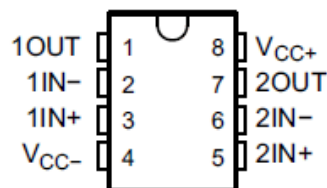
Electrical signals acquired from the human body are characterized by very low amplitudes. In the case of electroencephalography (EEG), the signal amplitude measured on the scalp using surface electrodes is typically in the range of a few microvolts (μV). Therefore, significant amplification is required before the signal can be digitized and visualized.

In this circuit, amplification is achieved through multiple cascaded stages.

Operational Amplifiers

The operational amplifiers used in this circuit are TL072 dual JFET-input op-amps. They were selected due to their high input impedance, low noise characteristics, and suitability for low-level bio-signal amplification. These properties make them appropriate for EEG signal conditioning stages following the instrumentation amplifier.

$$\text{Gain} = 1 + \frac{R_G}{R_F}$$



Functional Diagram of TL072 Op-amp

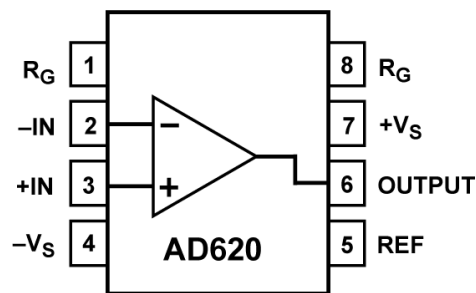
Instrumentation Amplifier

An instrumentation amplifier is placed directly after the electrodes to provide an initial amplification of the EEG signal while rejecting common-mode interference. The AD620 instrumentation amplifier is used for this purpose.

The gain of the AD620 is set by a single external resistor R_G , according to the manufacturer's datasheet:

$$Gain = 1 + \frac{49.4k\Omega}{R_G}$$

This stage amplifies the microvolt-level EEG signal to a level suitable for subsequent analog processing.



*Functional Diagram of AD620
Instrumentation Amplifier*

Total Gain

The total gain of the system is the product of the gains of all amplification stages:

$$Total\ Gain = G_{IA} \times G_{U2A} \times G_{U2B}$$

The resulting overall gain brings the EEG signal into a voltage range compatible with the ADS1115 analog-to-digital converter, enabling accurate digitization using an Arduino and real-time visualization in MATLAB.

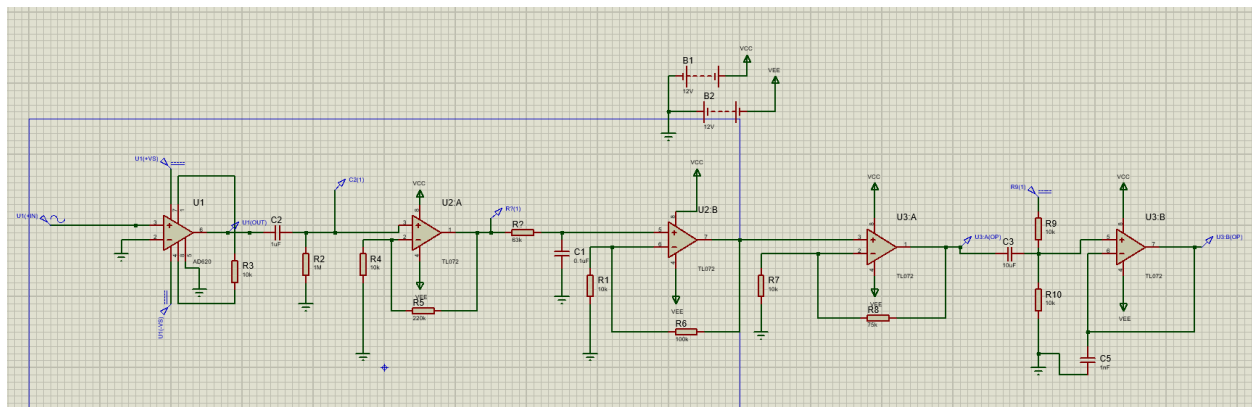
Complete Circuit For 1 Channel

To evaluate the behavior of the complete single-channel EEG acquisition circuit, a transient analysis was performed in order to observe the time-domain response of the system. This type of analysis allows the visualization of the signal evolution over time, similarly to an oscilloscope, and is well suited for verifying amplification and signal conditioning stages.

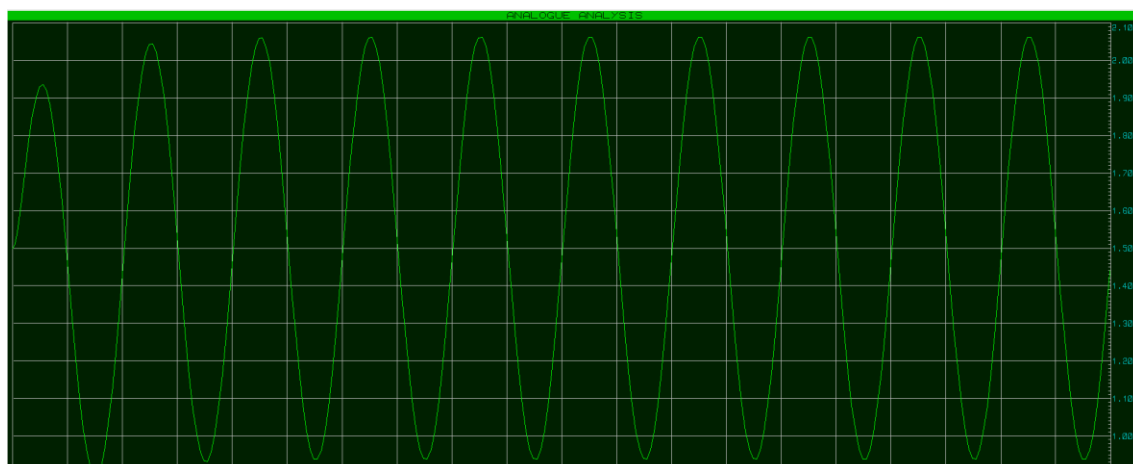
The input signal applied to the circuit is a sinusoidal waveform with a frequency of 10 Hz and an amplitude of 50 microvolt, representative of a typical EEG signal. After passing through the instrumentation amplifier, filtering stages, and amplification stages, the output signal is amplified to a voltage level suitable for analog-to-digital conversion.

As shown in the corresponding simulation results, the output signal preserves the original waveform while exhibiting the expected overall gain. Any DC offset introduced by the amplification stages is compensated at the final stage of the circuit using an operational amplifier combined with a voltage divider, ensuring that the output signal is properly biased within the acceptable input range of the ADC.

This final conditioning stage guarantees that the signal amplitude and offset are compatible with the ADS1115 input requirements, allowing accurate digitization and subsequent visualization.



Proteus Simulation of 1channel Circuit



Analogue Graph of 1channel Output

Other Components

Power Supply

The EEG acquisition circuit is powered using two 9V batteries to generate a dual supply of $\pm 9V$. This dual supply configuration is required to ensure proper operation of the AD620 instrumentation amplifier within its recommended input and output voltage ranges, and to prevent saturation when amplifying low-amplitude EEG signals. The same $\pm 9V$ supply rails are also used to power the operational amplifiers in the subsequent amplification and filtering stages.

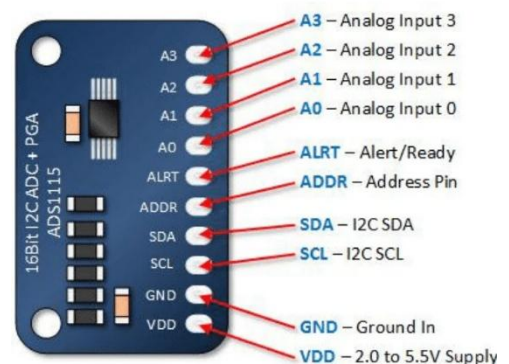
In addition, a separate low-voltage supply is required for the DC biasing stage at the output of the circuit. This stage is powered using a 3V supply provided directly by the Arduino. The DC bias circuit shifts the amplified EEG signal to the appropriate voltage level, ensuring compatibility with the input range of the ADC and allowing safe and accurate digitization.

This power supply configuration allows stable operation of all analog stages while maintaining correct signal centering for the data acquisition system.

ADC

The conditioned EEG signal is digitized using the ADS1115 analog-to-digital converter. The ADS1115 operates with a single-ended input voltage range from 0V to 3V and therefore cannot process bipolar signals centered around 0V.

The ADS1115 is interfaced with an Arduino, where the input EEG signal is read and transmitted to MATLAB code for real-time visualization.



Functioning Diagram of ADS1115 ADC

MATLAB & Arduino Codes:

Arduino Code

```
#include <Wire.h>
#include <Adafruit_ADS1X15.h>

Adafruit_ADS1115 ads;

void setup() {
  Serial.begin(9600);
  Wire.begin();
  if (!ads.begin()) {
    Serial.println("Failed to find ADS1115!");
    while (1);
  }
  ads.setGain(GAIN_ONE); //  $\pm 4.096\text{V}$ 
}

void loop() {
  int16_t value = ads.readADC_SingleEnded(1);
  float voltage = value * 0.125 / 1000; // GAIN_ONE: 0.125 mV per bit
  Serial.print("Raw: "); Serial.print(value);
  Serial.print("\tVoltage: "); Serial.println(voltage);
  delay(100);
}
```

MATLAB Code:

```
%% MATLAB real-time plot for ADS1115
clc; clear; close all;

% Serial port and settings
port = "/dev/cu.usbmodem214101"; % Arduino port (macOS)
baud = 9600; % must match Arduino Serial.begin()
fs = 250; % sampling frequency (Hz)
window = 5; % plot window in seconds
bufferLength = fs * window; % number of samples in window

% Close any existing serial connections on this port
existingPorts = serialportlist("available");
if any(existingPorts == port)
  % If port is already open in MATLAB, close it
  warning('Port is already in use. Make sure no other session is using it.');
```

```
end

% Open serial port
s = serialport(port, baud);
```

```

% Configure terminator for reading lines
configureTerminator(s,"LF");

% Pre-allocate data buffer
voltageBuffer = zeros(1, bufferLength);
timeBuffer = linspace(-window, 0, bufferLength);

% Create figure for real-time plot
hFig = figure('Name','ADS1115 Real-Time Plot');
hLine = plot(timeBuffer, voltageBuffer);
xlabel('Time (s)');
ylabel('Voltage (V)');
title('ADS1115 Channel 1');
ylim([0 5]); % adjust based on expected voltage range
grid on;

% Start real-time loop
disp('Starting real-time plotting...');
while ishandle(hFig)
    try
        % Read one line from Arduino
        line = readline(s);

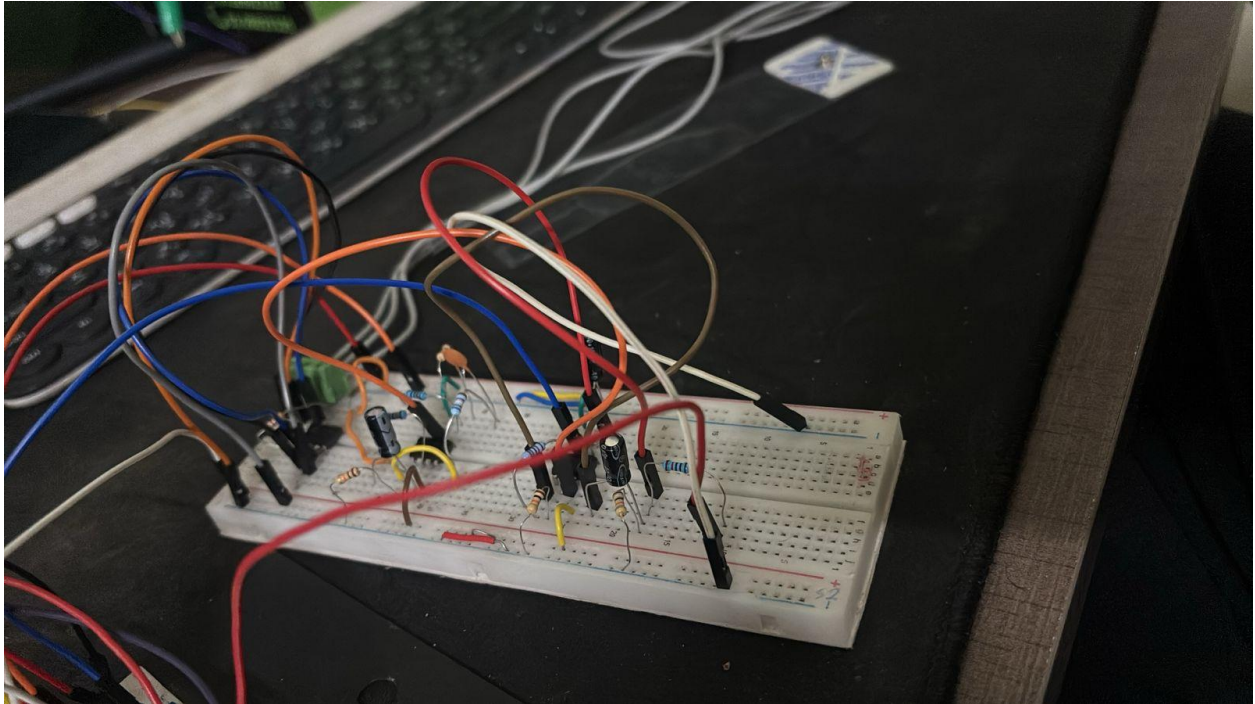
        % Parse the voltage from the line
        % Example Arduino line: "Raw: 4321 Voltage: 0.54"
        tokens = split(line);
        idx = find(strcmp(tokens,"Voltage:"),1);
        if ~isempty(idx)
            voltage = str2double(tokens{idx+1});

            % Shift buffer and append new value
            voltageBuffer = [voltageBuffer(2:end), voltage];

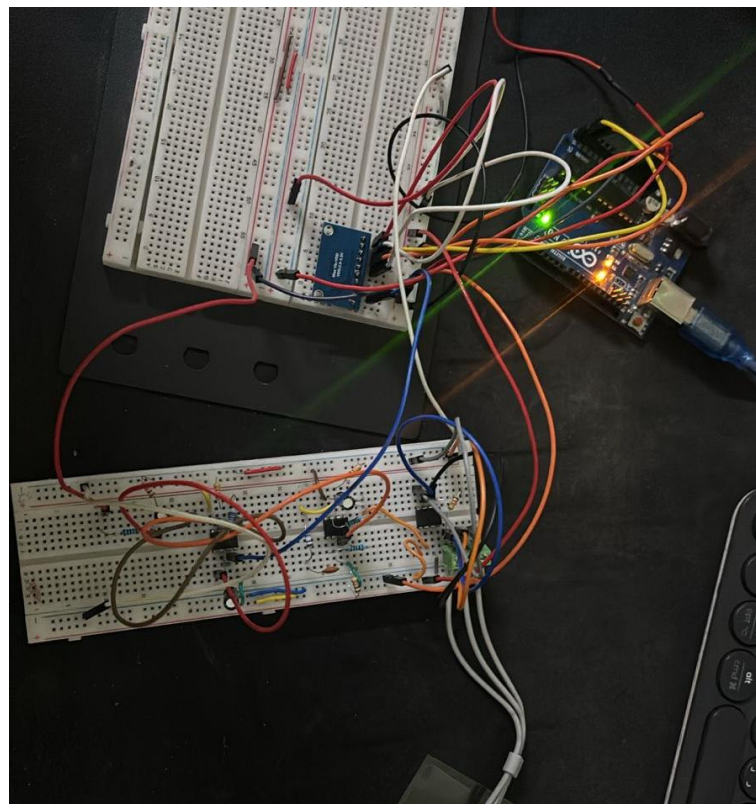
            % Update plot
            set(hLine,'YData',voltageBuffer);
            drawnow limitrate;
        end
    catch ME
        warning('Serial read error: %s', ME.message);
    end
end

% Close serial port when figure is closed
clear s;
disp('Serial port closed.');
```

Implemented EEG Acquisition Circuit



Implementation of 1 Channel



1 Channel Connected to ADC and Arduino

Board Cost:

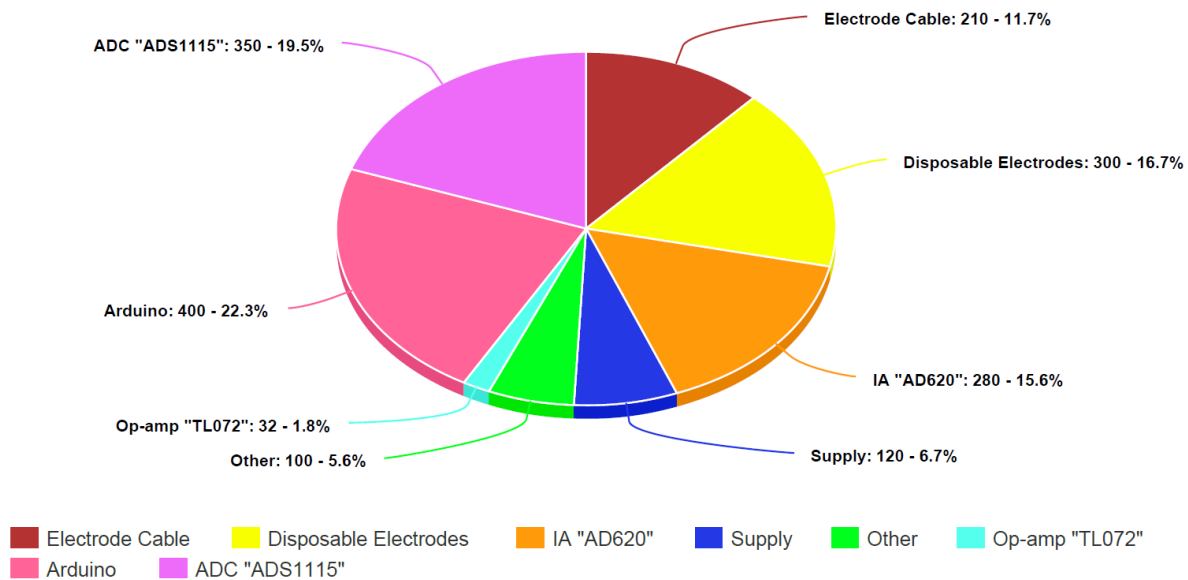
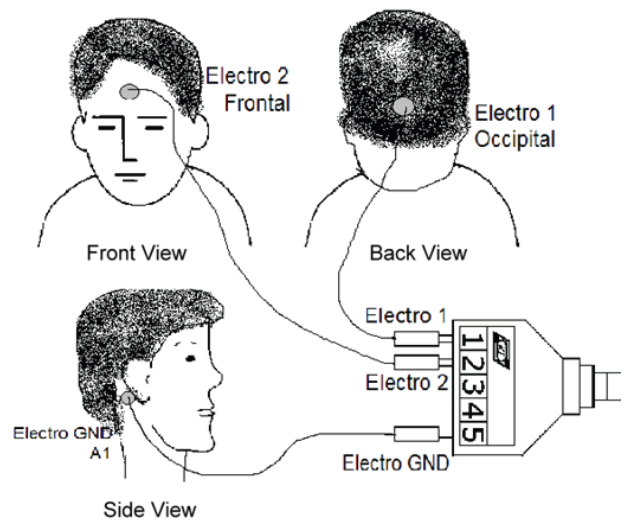


Chart: Board Cost

EEG Electrode Placement

EEG signal acquisition was performed using three surface electrodes placed at specific locations on the scalp. One electrode was positioned behind the right ear and used as a reference electrode. The second electrode was placed on the posterior region of the scalp to capture brain activity originating from the occipital area, while the third electrode was positioned on the frontal region of the scalp to record activity associated with the frontal lobe.

This electrode configuration allows the acquisition of EEG signals from different functional areas of the brain while maintaining a stable reference point. The selected placement provides clear voltage differences between the electrodes, enabling the observation of variations in brain activity between relaxed and active mental states.

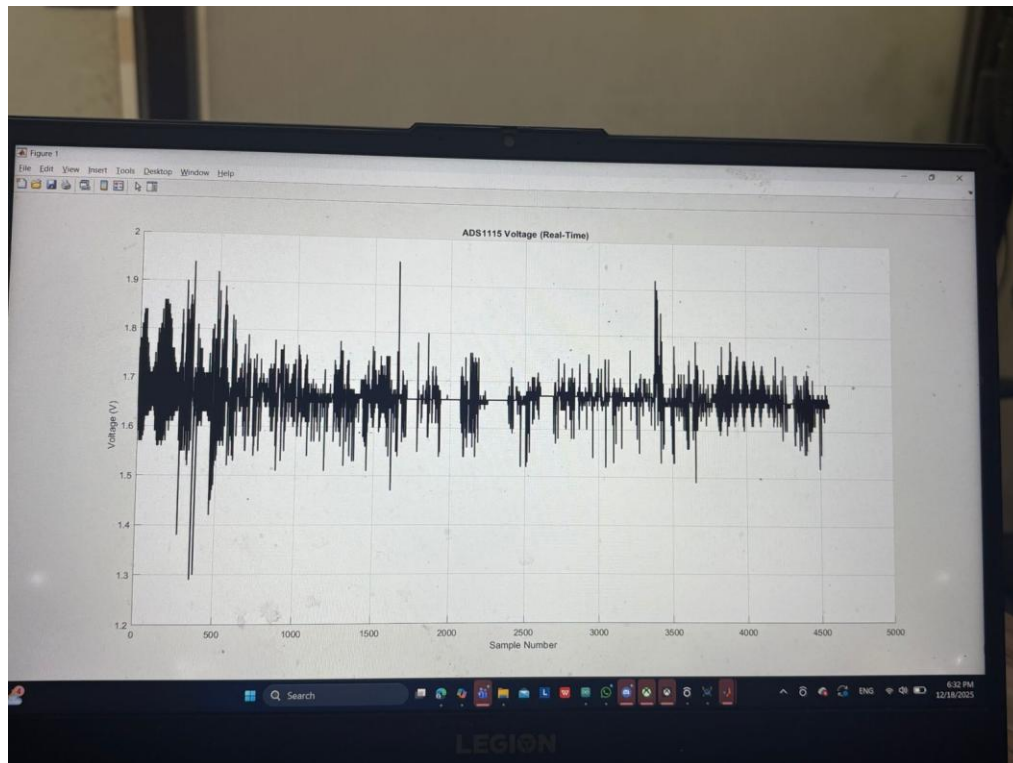


Electrodes Placement

Project Testing & Resulted Signal

The complete single-channel EEG acquisition circuit was tested multiple times to verify its performance and reliability. During testing, EEG data was recorded and transmitted to an Arduino, then streamed to MATLAB for real-time visualization.

The system was evaluated on multiple subjects to assess signal consistency and robustness. Despite the presence of environmental and physiological noise, the acquired EEG signals were clearly observable, and distinct variations could be identified. Changes in brain activity, such as transitions from a relaxed state to an active or focused state, resulted in noticeable differences in the recorded signal. These variations were consistently observed across multiple trials, confirming the correct operation of the acquisition, amplification, and digitization stages.



MATLAB Visualization in Real-Time

User Interface Implementation and Functionality

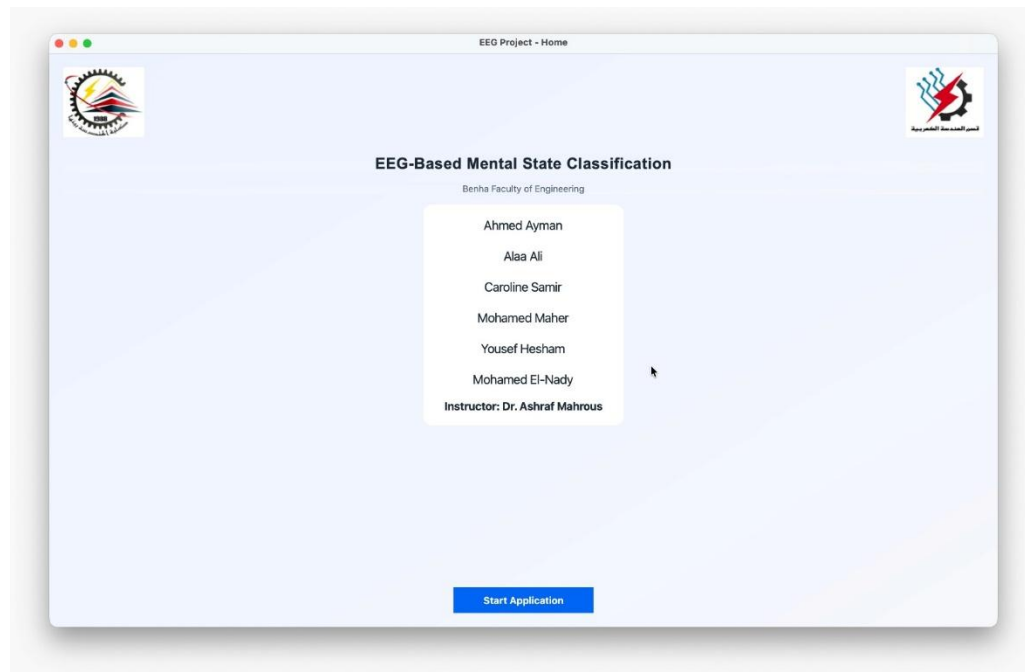
In order to facilitate real-time monitoring and analysis of EEG signals, a dedicated graphical user interface (GUI) was developed. The GUI serves as an interactive platform that connects to the EEG acquisition system and provides multiple functionalities for both live data visualization and offline analysis.

The interface displays the EEG signal in real-time, allowing the user to observe the brain activity as it occurs. This capability is particularly useful for immediate assessment of the subject's neurological state and for validating the performance of the acquisition circuit and amplifier stages. In addition to live signals, the GUI also supports playback of previously recorded EEG data. This feature enables comparative analysis and facilitates the study of patterns over time.

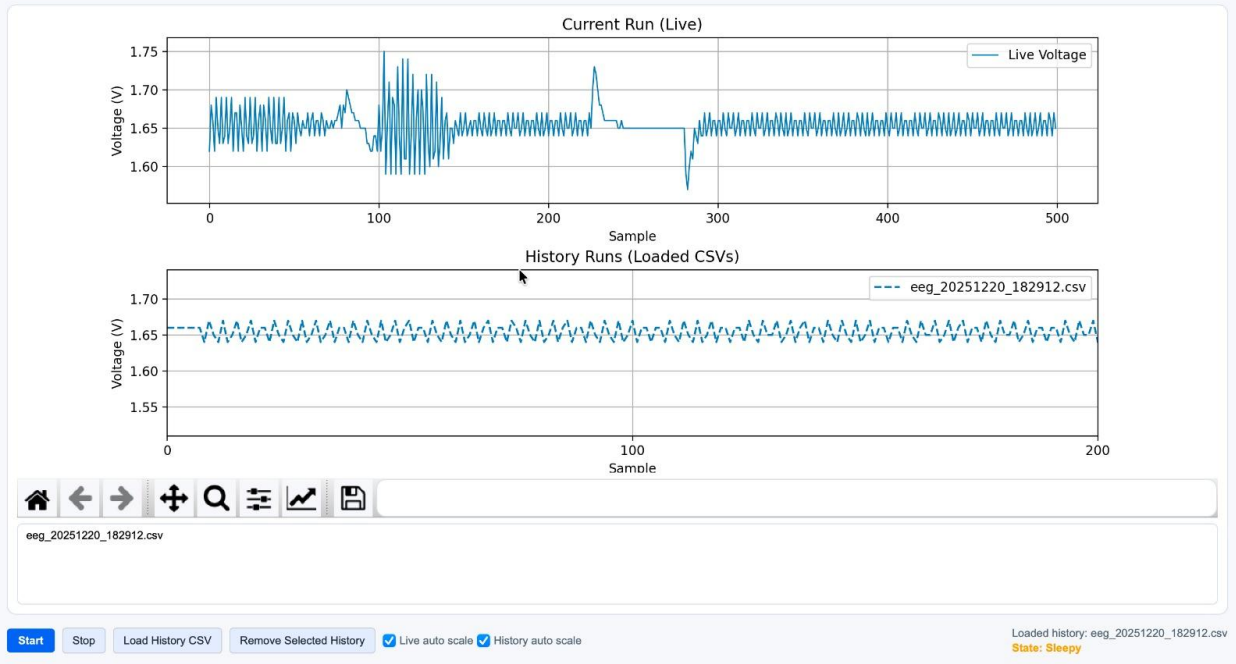
Beyond signal visualization, the GUI incorporates a state-monitoring feature that interprets the EEG data to estimate the current physiological or mental state of the subject. Using pre-defined criteria based on the amplitude and frequency characteristics of the signal. This feature provides an intuitive way to understand the relationship between EEG patterns and the subject's condition.

Although the underlying code for the GUI is extensive and not included in this report, it was designed with user-friendliness in mind, allowing operators to easily observe live and recorded data, and observe the real-time evaluation of the subject's state. This integration of signal acquisition, visualization, and interpretation demonstrates the practical application of EEG in a real-time monitoring context, bridging the gap between raw data and actionable insights.

The following figure shows the implemented user interface:



Graphical User Interface displaying real-time EEG signal at startup



EEG graphical user interface, showing the signal visualization

Applications of the EEG System

The implemented EEG acquisition system has several potential applications in both research and practical contexts. It can be used for monitoring brain activity in educational and experimental settings, allowing students and researchers to study neural patterns in real-time. The system is also suitable for evaluating the subject's physiological or mental state, such as detecting drowsiness, fatigue, or relaxation, which can be applied in cognitive research, ergonomics, and human-computer interaction studies.

Furthermore, the EEG system can serve as a foundation for more advanced applications, such as brain-computer interface (BCI) development, neurofeedback training, and clinical studies, where real-time monitoring and analysis of brain signals are essential. The modular design of the system makes it adaptable for integration with other sensors or data processing algorithms, enhancing its versatility for different experimental scenarios.

Encountered Difficulties

During the development and testing of our EEG acquisition system, several challenges were encountered that required careful consideration and problem-solving:

- **Lack of Clear References for Educational EEG Circuits:**

Most available resources focused on advanced EEG devices that are difficult to replicate in a small-scale educational setup. This lack of accessible references necessitated that we design the circuit from scratch, requiring significant analysis and understanding of each component to ensure proper functionality.

- **Unexpected Initial Measurements:**

Early tests of the circuit yielded unusual readings. Upon investigation, we discovered that the instrumentation amplifier (AD620) requires a dual supply voltage of ± 9 V in order to accurately amplify signals within the typical range of brain activity. The absence of the correct supply initially led to distorted measurements.

- **ADC Input Range Limitation:**

The analog-to-digital converter (ADC) used in our system had an input voltage range of 0–3.3 V. This limitation required the addition of a DC bias at the end of the amplification stage, shifting the EEG signal upward to fit within the ADC's measurable range while preserving the signal's waveform characteristics.

- **Design Iterations and Signal Optimization:**

Achieving stable and accurate signal acquisition required multiple iterations of the circuit and testing with real-time data. Fine-tuning the amplifier gain, filter settings, and bias voltage was essential to ensure the EEG signal could be reliably captured and visualized.